

Evaluation

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CS 382

Section 1

This table is basically for showcasing that the AI agent retrieves the right answers and hallucinates or not.

#	Query	Generated result	Notes
1	What is Laravel used for?	Laravel is a full-featured, open-source web application framework written in PHP that is designed around the principle of convention over configuration, and is used for building web applications and APIs in PHP	Single topic factual lookup
2	How do I define a model with GORM?	Based on the provided documents, GORM models are defined using normal/ordinary Go structs. These structs can contain fields with basic Go types, pointers or aliases of these types, or custom types (as long as they implement the Scanner and Valuer interfaces from the database/sql package). You can also embed `gorm.Model` directly into your structs to include common predefined fields. Column behavior can be expressed through struct tags.	Single topic factual lookup
3	How does routing work in Laravel?	Based on the provided documents, the most basic Laravel routes accept a URI and a closure, providing a method of defining routes and behavior without complicated routing configuration files: use Illuminate\Support\Facades\Route; Route::get('/greeting', function () { return 'Hello World'; });	Single topic factual lookup

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		All Laravel routes are defined in your route files, located in the routes directory, which are automatically loaded by Laravel using the configuration specified in your application's bootstrap/app.php file. The routes/web.php file defines routes for your web interface, which are assigned the web middleware group to provide features like session state and CSRF protection.	
4	How do I mock a repository dependency in a Spring Boot unit test?	To mock a repository dependency in a pure unit test where no Spring context is involved, you can use <code>@Mock</code> (from Mockito) to create a test double for the repository, and <code>@InjectMocks</code> to create a real instance of the class under test and inject the mock into it (as seen in the <code>OrderServiceTest</code> example).	Specific technique, needs precise term match
5	What's the difference between a stub and a mock?	Based on the provided documents, a stub returns hardcoded, canned responses to calls made during the test, without any real logic behind them. A mock is like a stub, but it also records how it was called (which methods, how many times, with what arguments) so the test can assert that specific interactions happened.	Specific terminology, could be confused with other testing docs
6	How do I prevent SQL injection when accepting user input?	To prevent SQL injection when accepting user input, you should use parameterized queries/prepared statements (or ORMs) so that user input is treated as data, not code.	Go between two docs, phrasing doesn't name either doc directly

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7	How does testing differ between Go, Laravel, and Spring Boot?	Based on the provided documents, testing differs across Go, Laravel, and Spring Boot in the following ways: Default Tooling and Philosophy: Go: Relies solely on the standard library's testing package with no external dependencies needed for basic unit tests. Table-driven tests and go test are idiomatic. Its testing story favors minimalism, where mocking relies on interfaces, and there are no built-in database-testing helpers (developers build their own conventions). (Source: Testing Comparison) Laravel: Provides a "batteries-included" and turnkey experience. It uses PHPUnit (or Pest) plus Laravel's own TestCase subclass pre-wired into every new project, adding HTTP simulation, database assertions, and facade faking. (Source: Testing Comparison, Laravel Testing) Spring Boot: Bundles JUnit 5, Mockito, and AssertJ into the single spring-boot-starter-test dependency. Spring Test adds framework-aware annotations (@SpringBootTest, @WebMvcTest, @DataJpaTest) for different testing scopes. (Source: Testing Comparison) Running Tests: Go: go test ./... (all packages), go test -v (verbose), or go test -cover (coverage). (Source: Testing Comparison) Laravel: php artisan test,	Requires synthesis across three ecosystems, easy to retrieve only one framework's doc instead of the comparison

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		vendor/bin/phpunit, or vendor/bin/pest. (Source: Testing Comparison) Spring Boot: mvn test or ./gradlew test. (Source: Testing Comparison) Test File Organization: Go: Tests live alongside the code they test in the same package and directory, with a _test.go suffix. (Source: Testing Comparison) Laravel: Tests live in a separate tests/ tree, split into Unit and Feature subdirectories. (Source: Testing Comparison) Spring Boot: Tests live in a parallel src/test/java tree that mirrors the main source package structure. (Source: Testing Comparison) Mocking: Go: Leans on the language's structural typing (interfaces) instead of a mocking framework. (Source: Testing Comparison) Laravel: Provides dedicated mocking utilities via facade calls (e.g., Mail::fake()) that cover framework-provided side effects, alongside PHPUnit-style mocks for custom classes. (Source: Testing Comparison) Spring Boot: Uses Mockito's @Mock and @InjectMocks for plain unit tests, and @MockBean to swap a bean inside a running Spring context for slice tests. (Source: Testing Comparison) Database Testing:	

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		Go: Has no built-in database test helpers; common patterns include an in-memory SQLite database, a Dockerized test database via testcontainers-go, or an in-memory fake repository. (Source: Testing Comparison) Laravel: Uses the RefreshDatabase trait, which wraps each test in a transaction or re-migrates to give every test a clean database automatically with one line of code. (Source: Testing Comparison) Spring Boot: Uses @DataJpaTest to boot only the JPA layer against an embedded H2 database by default, while Testcontainers is used when tests need to match production database behavior exactly. (Source: Testing Comparison)	
8	What's the best way to bake a sourdough loaf?	No relevant passages were found for that query.	Should show error message, not hallucinate an answer

While for the performance if we use the Gemini model all of those queries above took less than 5s to generate a proper response, but on the other hand using Ollama the local model will result in significantly higher time used approximately around 20 to 30s.